Kyra Kakar
Roll No:-2401038
AI and DS
Division:T1
Batch:T12
Subject: ASD & DevOps

EXPERIMENT:-1

Aim: To perform version control operations using Git and understand various Git commands for repository, file, branch, and commit management.

This experiment maps to LO2.

LO2: Use Git effectively for version control, source code management and collaborative team management.

Theory:

Git is a distributed version control system used to track changes in files and source code. It allows users to create repositories, maintain different versions of a project, work with branches, and view the history of changes. Git provides commands for adding, committing, comparing, merging, and managing files efficiently. In this experiment, 25 basic Git operations are performed to understand version control.

Commands:

No.	Command	Purpose
1	git --version	Checks the installed Git version.
2	git config --global user.name "Kyra Kakar"	Sets the Git username globally.
3	git config --global user.email "email"	Sets the Git email globally.
4	git config --list	Displays Git configuration details.
5	mkdir gitproject	Creates a new project folder.

Kyra Kakar
Roll No:-2401038
AI and DS
Division:T1
Batch:T12
Subject: ASD & DevOps

6	cd gitproject	Enters the project folder.
7	git init	Initializes a new Git repository.
8	echo "Hello Git" > file.txt	Creates a new file with content.
9	cat file.txt	Displays the contents of the file.
10	git status	Checks the current repository status.
11	git add file.txt	Adds a specific file to the staging area.
12	git add .	Adds all modified files to the staging area.
13	git commit -m "Initial commit"	Creates the first committed version.
14	git log	Displays detailed commit/version history.
15	git log --oneline	Displays different versions using short commit IDs.
16	echo "Git Version Control" >> file.txt	Adds content to an existing file.
17	git diff	Displays changes between the current and committed file.
18	git restore file.txt	Rolls back uncommitted file changes.

Kyra Kakar
Roll No:-2401038
AI and DS
Division:T1
Batch:T12
Subject: ASD & DevOps

19	git reset HEAD~1	Rolls back the latest commit while retaining changes.
20	git branch feature	Creates a new branch named feature.
21	git branch	Displays all branches and the current branch.
22	git checkout feature	Switches to the feature branch.
23	git switch feature	Switches to the feature branch using the modern Git command.
24	echo "Feature Branch" > feature.txt	Creates a file in the feature branch.
25	git checkout main	Switches back to the main branch.
26	git switch main	Switches back to the main branch using git switch.
27	git merge feature	Merges the feature branch with the current branch.
28	git branch -d feature	Deletes the feature branch.
29	git remote add origin <URL>	Connects the local repository to a remote repository.
30	git remote -v	Displays configured remote repositories.

Kyra Kakar
Roll No:-2401038
AI and DS
Division:T1
Batch:T12
Subject: ASD & DevOps

31	git push -u origin main	Pushes local commits to the remote repository and authenticates if required.
32	git pull origin main	Retrieves and merges changes from the remote repository.
33	git clone <URL>	Creates a local copy of a remote repository.
34	git show <commit-id>:file.txt	Checks the content of a file in a specific version.
35	git checkout <commit-id>	Switches the repository to a previous committed version.
36	git checkout main	Returns from an older version to the latest main branch.
37	git restore --source=<commit-id> file.txt	Restores a file from a specific older version.

Output:

```
kyra@Kyras-MacBook-Air ~ % git --version
git version 2.50.1 (Apple Git-155)
kyra@Kyras-MacBook-Air ~ % git config --global user.name "Kyra Kakar"
kyra@Kyras-MacBook-Air ~ % git config --global user.email "kyrakakar@gmail.com"
kyra@Kyras-MacBook-Air ~ % git config --list
credential.helper=osxkeychain
init.defaultbranch=main
user.name=Kyra Kakar
user.email=kyrakakar@gmail.com
kyra@Kyras-MacBook-Air ~ % cd ~/Desktop
kyra@Kyras-MacBook-Air Desktop % mkdir gitproject
kyra@Kyras-MacBook-Air Desktop % cd gitproject
kyra@Kyras-MacBook-Air gitproject % git init
Initialized empty Git repository in /Users/kyra/Desktop/gitproject/.git/
kyra@Kyras-MacBook-Air gitproject % echo "Hello Git" > file.txt
kyra@Kyras-MacBook-Air gitproject % cat file.txt
Hello Git
kyra@Kyras-MacBook-Air gitproject % git config --global user.email "kyrakakar@gmail.com"
kyra@Kyras-MacBook-Air gitproject % git add file.txt
kyra@Kyras-MacBook-Air gitproject % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   file.txt
```

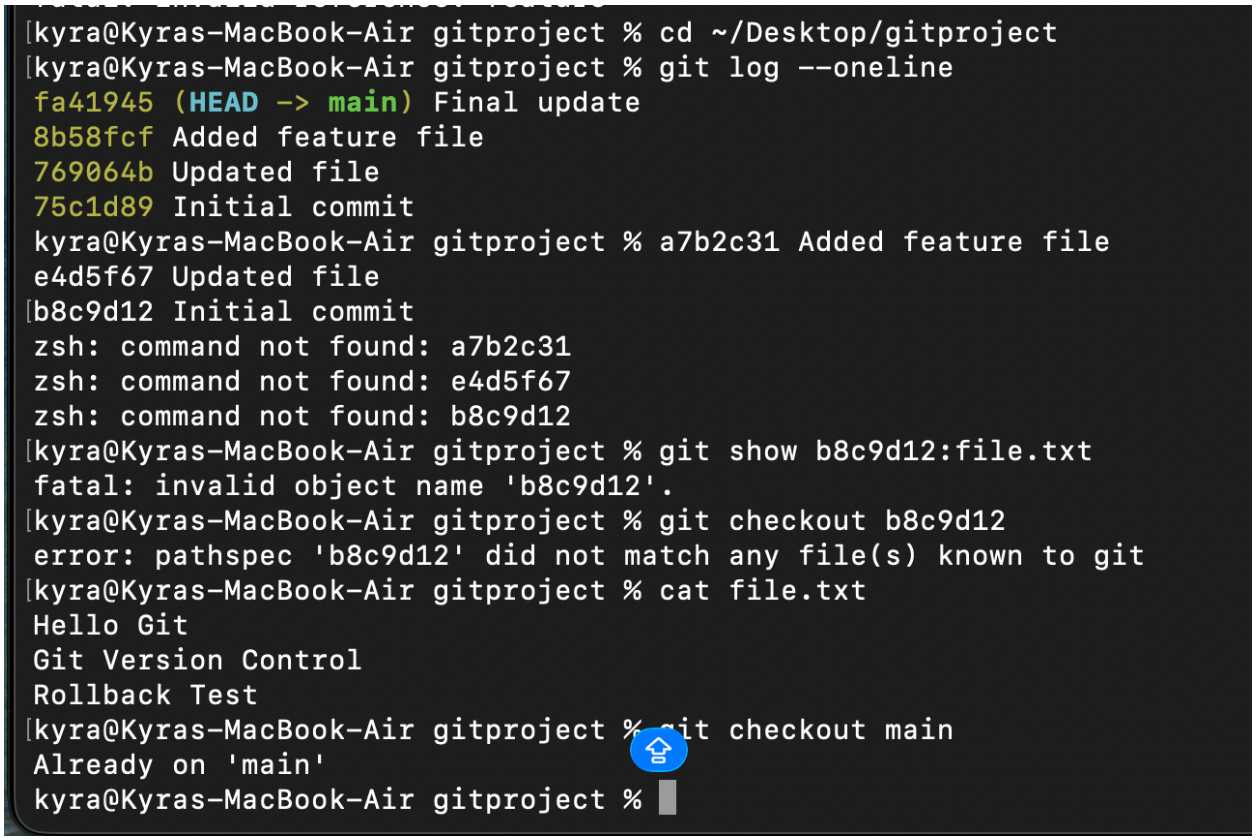
Kyra Kakar
Roll No:-2401038
AI and DS
Division:T1
Batch:T12
Subject: ASD & DevOps

```
* main
[kyra@Kyras-MacBook-Air gitproject % [200~cd ~/Desktop~
zsh: bad pattern: [200~cd
[kyra@Kyras-MacBook-Air gitproject % cd ~/Desktop
[kyra@Kyras-MacBook-Air Desktop % cd gitproject
[kyra@Kyras-MacBook-Air gitproject % echo "Temporary Change" >> file.txt
[kyra@Kyras-MacBook-Air gitproject % git diff
diff --git a/file.txt b/file.txt
index e4fc519..d3775d9 100644
--- a/file.txt
+++ b/file.txt
@@ -1,2 +1,3 @@
 Hello Git
 Git Version Control
+Temporary Change
[kyra@Kyras-MacBook-Air gitproject % git restore file.txt
[kyra@Kyras-MacBook-Air gitproject % cat file.txt
Hello Git
Git Version Control
kyra@Kyras-MacBook-Air gitproject % echo "Rollback Test" >> file.txt
git add .
[git commit -m "Rollback test"
[main 19f1b99] Rollback test
 1 file changed, 1 insertion(+)
[kyra@Kyras-MacBook-Air gitproject % git log --oneline
19f1b99 (HEAD -> main) Rollback test
8b58fcf Added feature file
769064b Updated file
75c1d89 Initial commit
[kyra@Kyras-MacBook-Air gitproject % git reset HEAD~1
Unstaged changes after reset:
M       file.txt
[kyra@Kyras-MacBook-Air gitproject % git log --oneline
8b58fcf (HEAD -> main) Added feature file
769064b Updated file
75c1d89 Initial commit
[kyra@Kyras-MacBook-Air gitproject % git remote add origin YOUR_REPOSITORY_URL
[kyra@Kyras-MacBook-Air gitproject % git remote -v
origin  YOUR_REPOSITORY_URL (fetch)
origin  YOUR_REPOSITORY_URL (push)
kyra@Kyras-MacBook-Air gitproject % git add .
[git commit -m "Final update"
[main fa41945] Final update
 1 file changed, 1 insertion(+)
[kyra@Kyras-MacBook-Air gitproject % git push -u origin main
fatal: 'YOUR_REPOSITORY_URL' does not appear to be a git repository
fatal: Could not read from remote repository.
```

Kyra Kakar
Roll No:-2401038
AI and DS
Division:T1
Batch:T12
Subject: ASD & DevOps

```
    Initial commit
[kyra@Kyras-MacBook-Air gitproject % git log --oneline
75c1d89 (HEAD -> main) Initial commit
[kyra@Kyras-MacBook-Air gitproject % echo "Git Version Control" >> file.txt
[kyra@Kyras-MacBook-Air gitproject % cat file.txt
Hello Git
Git Version Control
[kyra@Kyras-MacBook-Air gitproject % git diff
diff --git a/file.txt b/file.txt
index 9f4d96d..e4fc519 100644
--- a/file.txt
+++ b/file.txt
@@ -1 +1,2 @@
 Hello Git
+Git Version Control
[kyra@Kyras-MacBook-Air gitproject % git add .
[kyra@Kyras-MacBook-Air gitproject % git commit -m "Updated file"
[main 769064b] Updated file
 1 file changed, 1 insertion(+)
[kyra@Kyras-MacBook-Air gitproject % git branch feature
[kyra@Kyras-MacBook-Air gitproject % git branch
  feature
* main
[kyra@Kyras-MacBook-Air gitproject % git checkout feature
Switched to branch 'feature'
[kyra@Kyras-MacBook-Air gitproject % echo "Feature Branch" > feature.txt
[kyra@Kyras-MacBook-Air gitproject % git add feature.txt
[kyra@Kyras-MacBook-Air gitproject % git add feature.txt
[kyra@Kyras-MacBook-Air gitproject % git commit -m "Added feature file"
[feature 8b58fcf] Added feature file
 1 file changed, 1 insertion(+)
 create mode 100644 feature.txt
[kyra@Kyras-MacBook-Air gitproject % git branch
* feature
  main
[kyra@Kyras-MacBook-Air gitproject % git checkout main
Switched to branch 'main'
[kyra@Kyras-MacBook-Air gitproject % git merge feature
Updating 769064b..8b58fcf
Fast-forward
 feature.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 feature.txt
[kyra@Kyras-MacBook-Air gitproject % git branch -d feature
Deleted branch feature (was 8b58fcf).
[kyra@Kyras-MacBook-Air gitproject % git branch
* main
```

Kyra Kakar
Roll No:-2401038
AI and DS
Division:T1
Batch:T12
Subject: ASD & DevOps



Conclusion: Through this experiment, I learned how to use Git as a version control system and perform various operations for managing changes in a project. I understood how to initialize a Git repository, configure user details, track files, stage changes, and create commits to maintain different versions of a project. I also learned how to view commit history and compare modifications made to files. Furthermore, I gained practical knowledge of branch management by creating, switching, merging, and deleting branches. The experiment also helped me understand rollback operations using Git restore and reset commands to undo unwanted changes and commits. Additionally, I learned how to connect a local repository to a remote GitHub repository, authenticate and push changes, pull updates, and clone repositories. Overall, this experiment gave me a clear understanding of how Git helps in tracking changes, maintaining project history, recovering previous versions, and managing code efficiently during software development.